

- 5 An ideal transformer is connected to a sinusoidal a.c. supply, as shown in Fig. 5.1. The primary coil has a r.m.s. current of 0.85 A.

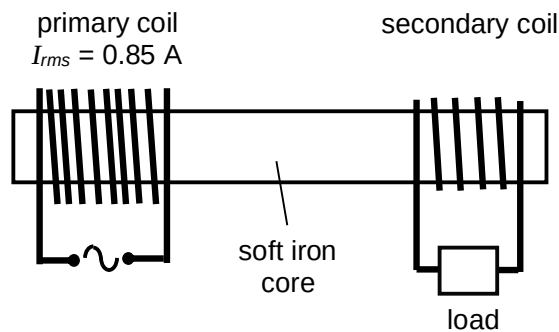


Fig. 5.1

- (a) Use the laws of electromagnetic induction to explain how a potential difference can be developed across the secondary coil.

..... Alternating current in the primary coil results in an alternating magnetic field
 strength generated. [M1]
 Magnetic field strength generated by primary coil linked to secondary coil (via the
 soft iron core).
 Secondary coil experiences alternating magnetic flux linkage due to alternating
 magnetic field strength [M1]. By Faraday's law, an e.m.f. will be induced across
 the secondary coil. [A1]
 Without 1st M1, max 1 mark.

[3]

- (b) (i) The primary coil contains 9 turns per cm, calculate the maximum magnetic field strength at its centre.

$$B = \mu_0 n I = 4\pi \times 10^{-7} \times 9 \times 100 \times 0.85\sqrt{2} = 1.36 \times 10^{-3} \text{ T} \quad [2]$$

maximum magnetic field strength = T [2]

- (ii) The ratio of the number of turns in the primary to secondary coil is 16 : 1.

Calculate the peak current in the load.

$$\frac{I_s}{I_p} = \frac{N_p}{N_s} = 16$$

$$I_s = 16 \times 0.85\sqrt{2} \quad [\text{M1}]$$

$$= 19.2 \text{ A} \quad [\text{A1}]$$

peak current = A [2]

- (iii) The variation with time t of current I_p in the primary coil is shown in Fig. 5.2.

On Fig. 5.3, sketch the variation with time t of current I_s in the secondary coil. [1]

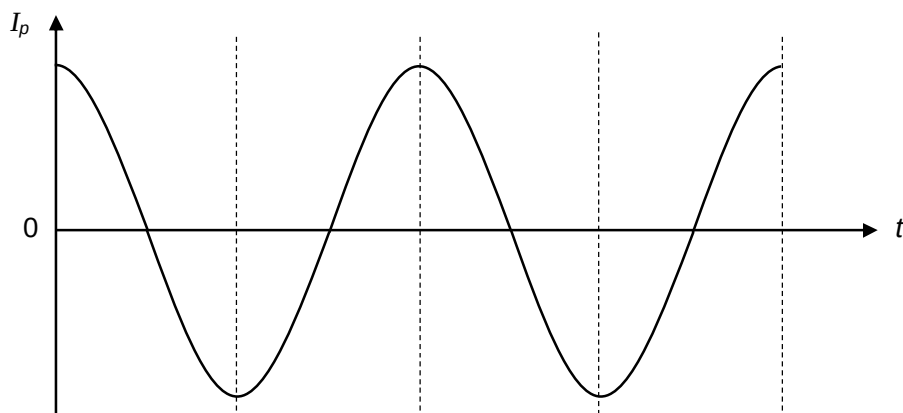


Fig. 5.2

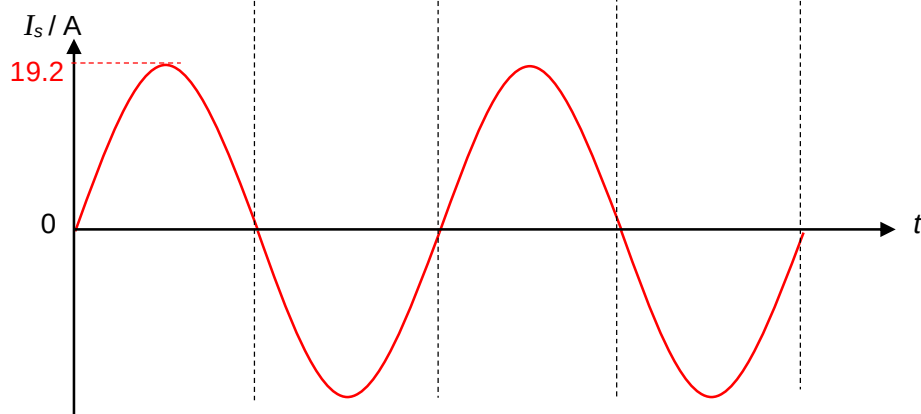


Fig. 5.3

- (iv) State and explain how the answer to b (iii) will change if the iron core is removed from the transformer.

Magnetic flux linking secondary coil will decrease. [M1]
Hence rate of change of flux linkage will decrease. (By Faraday's law, induced e.m.f. and) hence magnitude of induced current will decrease. [A1]

....

....

[2]

6 Ultraviolet radiation of wavelength 122 nm is used to illuminate the cathode in the vacuum